

Sayantani Bhattacharya

Department of Mathematics and Statistics, Auburn University, AL, 36849

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Education

Auburn University, Alabama, USA

Aug 2023 – Present

PhD in Mathematics [CGPA : 3.67]

Tezpur University, Assam, India

2018 – 2020

Master's in Mathematics

Chennai Mathematical Institute, Chennai, India

2015 – 2018

Bachelor's in Mathematics and Computer Science (Dual Degree)

Research Interests

Matroid Theory, Combinatorics.

Preprints & Publications

1. S. Bhattacharya, J. D. Clifton, Z. Walsh, The unbreakable quasi-graphic matroids. **Submitted.** [[arxiv](#)]

Talks

- ◇ Combinatorics Seminar, Auburn University, Auburn, AL (8th April, 2026)
The Unbreakable Quasi-graphic Matroids
- ◇ 34th Cumberland Conference on Combinatorics, Graph Theory and Computing, Auburn University, Auburn, AL (17th May, 2026)
The Unbreakable Quasi-graphic Matroids

Teaching Experience

Graduate Teaching Assistant, Auburn University, AL

<i>MATH 1610.R25, 1610.R26 (Calculus II) : Recitation Leader.</i>	Spring 2026
<i>Merit Recitation Program MATH 1950 MR1 (Calculus I) : Instructor of Record.</i>	Fall 2025
<i>Merit Recitation Program (Calculus I) : Graduate Research Assistant.</i>	Summer 2025
<i>MATH 1610.R15, 1610.R16 (Calculus I) : Recitation Leader.</i>	Spring 2025
<i>MATH 1000.005, 1000.006 (College Algebra) : Instructor of Record.</i>	Fall 2024
<i>MATH 1000/1120/30/50 & 2650 : GTA (Tutoring Center).</i>	Summer 2024
<i>MATH 1610, 1617 (Calculus I) : Grader.</i>	Spring 2024
<i>MATH 2650 (Linear Differential Equations) : Grader.</i>	Fall 2023

Graduate Research Assistant, Auburn University, AL

Research Assistantship under Dr. Zach Walsh for Summer 2026 under NSF Grant DMS-2452015.

Fellowships & Awards

- ◇ Awarded \$1000 towards Professional Development for the completion of the Auburn University Summer STEM Institute (AUSSI 2025) Training Program.
- ◇ Awarded \$4500 for a Graduate Research Assistantship position in the “Merit Recitation Program” at Auburn University during the Summer 2025. Awarded a Certificate of Completion for the “2025 Active Learning Teaching Professional Development Program” in Spring 2025.
- ◇ Graduate Teaching Assistantship and Fellowship, Auburn University (2023–Present)
- ◇ Council of Scientific and Industrial Research (CSIR India) National Eligibility Test (NET) with Junior Research Fellowship (JRF) Awardee (Nov 2019)
- ◇ INSPIRE Scholarship for Higher Education (SHE) Fellowship, Govt. of India (2015–2018)
- ◇ Institutional Fellowship at Chennai Mathematical Institute for Undergraduates (2015-2018)

Service & Outreach

- ◇ DMS Volunteer, **34th Cumberland Conference**, Auburn University (May 15-17th, 2026)
- ◇ Volunteer, **Mathematical Puzzle Programs (MaPP)**, Auburn University (Feb 21st, 2026)
- ◇ Volunteer, Bungee Drop Div C, **Alabama State Science Olympiad**, Auburn University (Feb 14th, 2026)
- ◇ Volunteer, **Julia Robinson Mathematics Festival (JRMF) at Joint Mathematics Meeting (JMM)**, Washington DC (Jan 4th, 2026)
- ◇ Exhibitor, **Destination STEM Expo**, Auburn University (Oct 15th, 2025)
- ◇ Volunteer, Codebusters Div B, **Alabama State Science Olympiad**, Auburn University (Mar 29th, 2025)
- ◇ Mentor, **First Year Experience (FYE) Mentorship Program**, Graduate Student Council, Auburn University (since Feb 2025)
 - ▷ *Awarded the Outstanding First Year Experience (FYE) Mentor Award 2025 by the Auburn University Graduate School.*
- ◇ Volunteer, **Mathematical Puzzle Programs (MaPP)**, Auburn University (Feb 15th, 2025)
- ◇ President, **College of Science & Mathematics International Students Association (CISA)**, Auburn University (Nov 2023 – Aug 2024)

Workshops & Summer Schools

- ◇ **Science Communication Workshop Series**, Biggio Center, Auburn University, Fall 2025
- ◇ **Topological Methods in Combinatorics Summer School**, Iowa State University, May 2025
- ◇ **Advanced Instructional School on Advanced Linear Algebra (AIS-ALA 2022)**, ATM Schools, National Center for Mathematics, Mumbai, India, May, 2022

Technical Skills

Programming: Python, C, SageMath, LaTeX